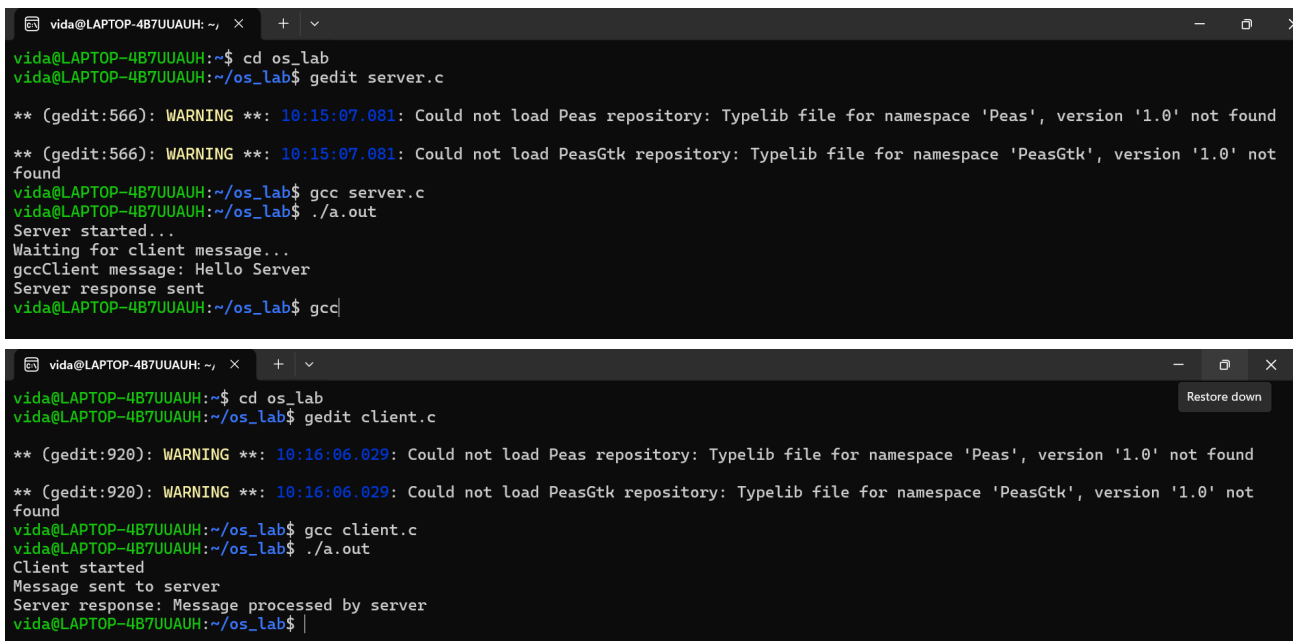


Practical-6

First Task: Client-Server Communication using Named Pipes (FIFOs)

- A client-server application was created using Named Pipes (FIFOs).
- The client sends a message to the server through the FIFO.
- The server receives the message, processes it and sends a response back.
- The server waits for a client message before processing it.
- The output showed successful communication between the client and server.
- When multiple clients communicate at the same time, the FIFO handles the messages in the order in which they are written to the pipe.

Screenshot of the Output



```
vida@LAPTOP-4B7UUAUH: ~$ cd os_lab
vida@LAPTOP-4B7UUAUH:~/os_lab$ gedit server.c

** (gedit:566): WARNING **: 10:15:07.081: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:566): WARNING **: 10:15:07.081: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc server.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Server started...
Waiting for client message...
gccClient message: Hello Server
Server response sent
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc
```

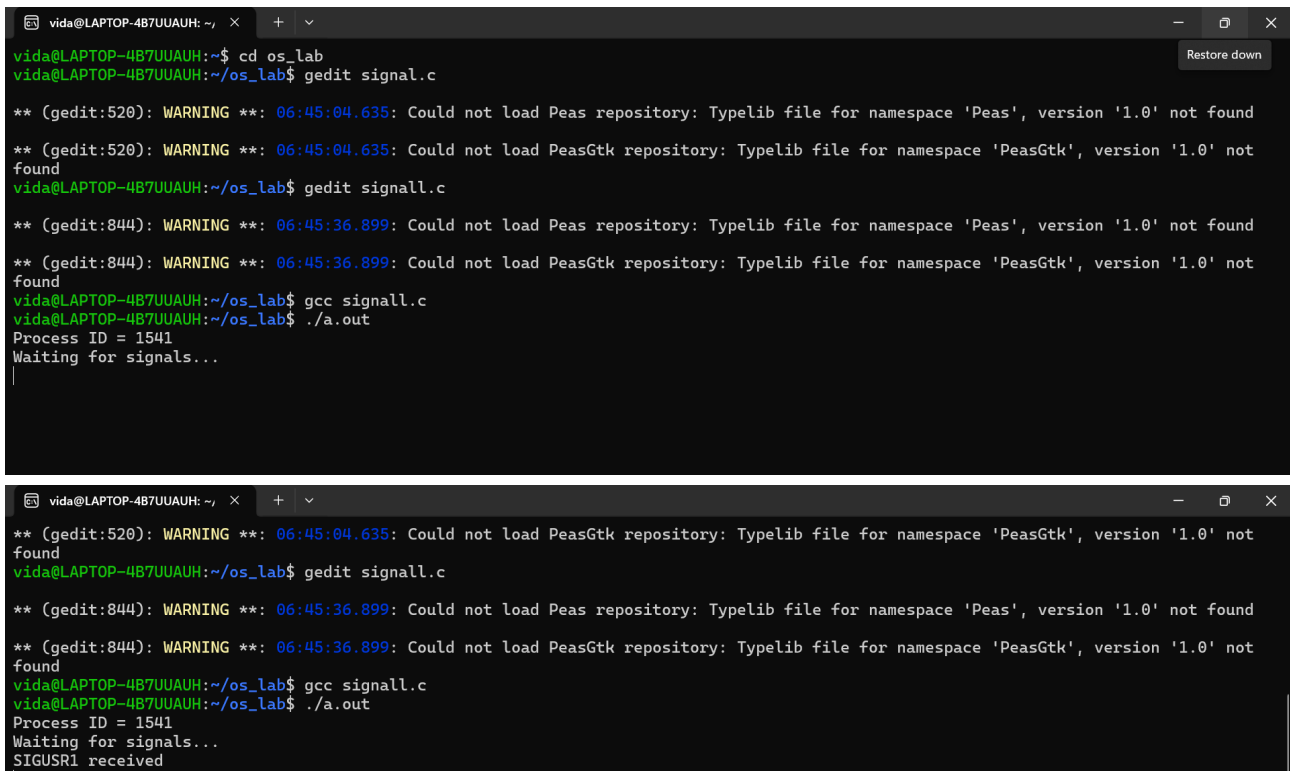
```
vida@LAPTOP-4B7UUAUH: ~$ cd os_lab
vida@LAPTOP-4B7UUAUH:~/os_lab$ gedit client.c

** (gedit:920): WARNING **: 10:16:06.029: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:920): WARNING **: 10:16:06.029: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc client.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Client started
Message sent to server
Server response: Message processed by server
vida@LAPTOP-4B7UUAUH:~/os_lab$
```

Second Task: POSIX Signal Handling

- A POSIX signal handling program was created to handle SIGINT, SIGTERM and SIGUSR1.
- The program displays its Process ID and waits for signals.
- SIGUSR1, SIGTERM and SIGINT were sent to the running process.
- A signal handler receives the signal and performs the required action.
- Signal handling is asynchronous because the signal can be received while the process is waiting or performing another task.

Screenshot of the Output



```
vida@LAPTOP-4B7UUAUH: ~/os_lab
vida@LAPTOP-4B7UUAUH:~/os_lab$ gedit signal.c

** (gedit:520): WARNING **: 06:45:04.635: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:520): WARNING **: 06:45:04.635: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gedit signall.c

** (gedit:844): WARNING **: 06:45:36.899: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:844): WARNING **: 06:45:36.899: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc signall.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Process ID = 1541
Waiting for signals...

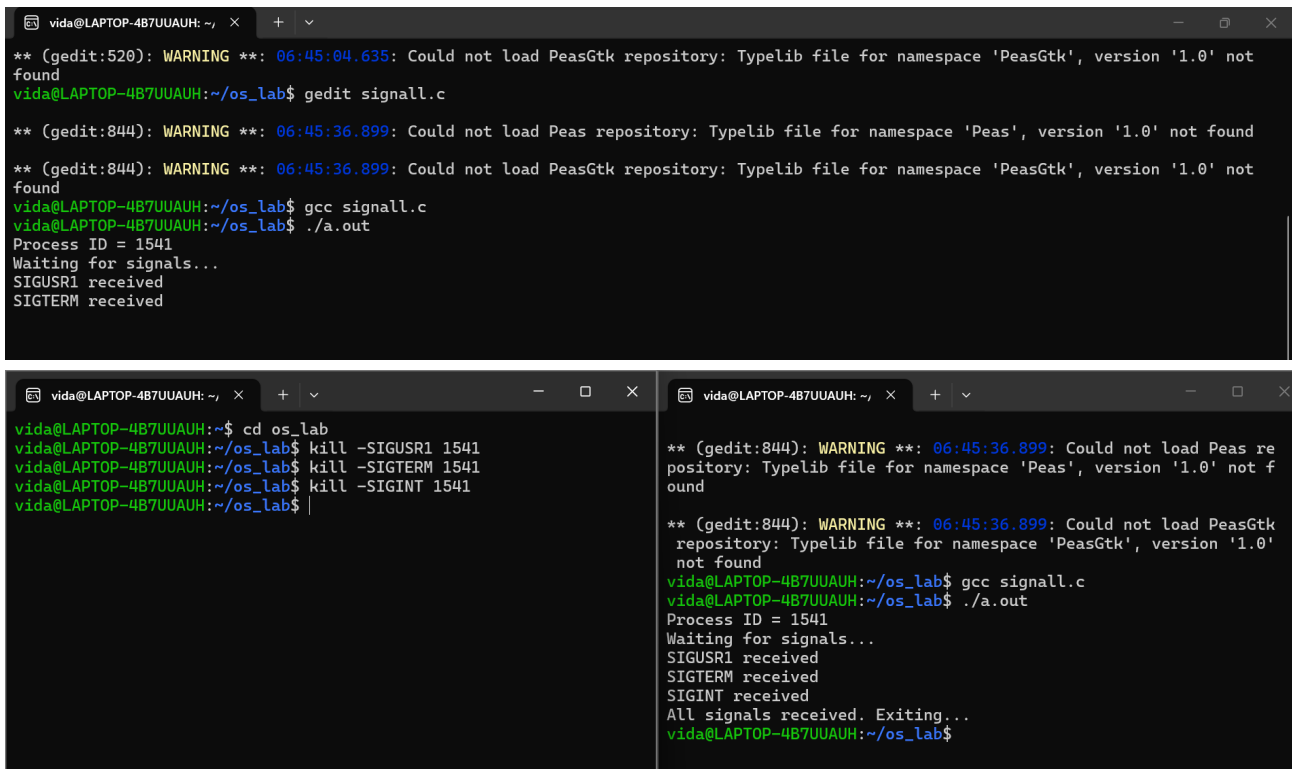
** (gedit:520): WARNING **: 06:45:04.635: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gedit signall.c

** (gedit:844): WARNING **: 06:45:36.899: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:844): WARNING **: 06:45:36.899: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc signall.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Process ID = 1541
Waiting for signals...
SIGUSR1 received
```

Second Task: POSIX Signal Handling

- SIGTERM was received and handled by the program.
- SIGINT was also received and handled successfully.
- After all required signals were received, the program displayed the exit message.
- The signal handler allows the process to respond to signals in a controlled way.

Screenshot of the Execution Output



```
vida@LAPTOP-4B7UUAUH: ~/os_lab$ gedit signall.c
** (gedit:520): WARNING **: 06:45:04.635: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc signall.c
** (gedit:844): WARNING **: 06:45:36.899: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:844): WARNING **: 06:45:36.899: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Process ID = 1541
Waiting for signals...
SIGUSR1 received
SIGTERM received

vida@LAPTOP-4B7UUAUH:~/os_lab$ cd os_lab
vida@LAPTOP-4B7UUAUH:~/os_lab$ kill -SIGUSR1 1541
vida@LAPTOP-4B7UUAUH:~/os_lab$ kill -SIGTERM 1541
vida@LAPTOP-4B7UUAUH:~/os_lab$ kill -SIGINT 1541
vida@LAPTOP-4B7UUAUH:~/os_lab$
```

```
** (gedit:844): WARNING **: 06:45:36.899: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:844): WARNING **: 06:45:36.899: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
vida@LAPTOP-4B7UUAUH:~/os_lab$ gcc signall.c
vida@LAPTOP-4B7UUAUH:~/os_lab$ ./a.out
Process ID = 1541
Waiting for signals...
SIGUSR1 received
SIGTERM received
SIGINT received
All signals received. Exiting...
vida@LAPTOP-4B7UUAUH:~/os_lab$
```

Result: Both client-server communication using FIFO and POSIX signal handling were successfully implemented and executed.